Characterizing Sensor Drift and Cross-Calibration in Korea's National Air Quality Monitoring Network

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Abstract

We present a comprehensive empirical characterization of sensor performance across 350 stations in Korea's national air quality monitoring network (AirKorea), analyzing 748,792 hourly measurements of six pollutants over a 89-day period (6 December 2025–6 March 2026). Our analysis reveals statistically significant drift signals, consistent with possible sensor drift, in 33.8% of $PM_{2.5}$ stations and up to 61.3% of SO_2 stations (linear trend $p < 0.05$), with drift rates varying substantially across pollutant types. Spatial cross-calibration of 279 adjacent station pairs (<5 km) yields a median Pearson correlation of 0.936 for $PM_{2.5}$, but 23.7% of pairs exhibit systematic biases exceeding $5\,\mu g/m^3$. We also document pronounced heteroscedastic noise characteristics: daily coefficient of variation (CV) ranges from 0.115 (SO_2) to 0.365 (NO_2/O_3), with CV inversely related to concentration level. Station failure analysis reveals 36.3% of stations experienced data gaps exceeding 24 hours, with mean time between failures of 192.8 days for urban stations. These findings provide actionable benchmarks for calibration scheduling, quality control, and sensor network design. We propose a Sensor Health Index (SHI) that integrates drift, noise, and data availability into a single composite metric for prioritizing station maintenance.

Keywords: Air quality monitoring, sensor drift, cross-calibration, $PM_{2.5}$, network characterization

1 Introduction

Air quality monitoring networks are critical infrastructure for public health protection and environmental policy. Korea operates one of the world's densest national monitoring networks through AirKorea, comprising over 670 stations that continuously measure six criteria pollutants: $PM_{2.5}$, PM_{10}, SO_2, NO_2, CO, and O_3 [1]. While these stations employ reference-grade analyzers (beta attenuation monitors for PM, UV fluorescence for SO_2, chemiluminescence for NO_2), systematic characterization of their in-field performance—including drift, noise, and inter-station consistency—remains limited.

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Prior studies have examined sensor performance in controlled laboratory settings [2] or focused on low-cost sensor calibration [3, 4]. However, reference-grade instruments deployed in national networks also exhibit drift, calibration offsets, and failure modes that can compromise data quality [5]. Understanding these characteristics at network scale is essential for: (1) optimizing calibration schedules, (2) detecting malfunctioning stations, and (3) establishing uncertainty bounds for epidemiological and regulatory applications.

While three months may appear limited for long-term drift assessment, this period represents a standard quarterly calibration interval—the minimum period over which drift should be detectable if calibration schedules are adequate.

Despite the importance of holistic sensor performance assessment, no standardized composite metric exists for integrating multiple degradation indicators into actionable maintenance priorities.

In this paper, we present, to our knowledge, the first systematic empirical characterization of in-field sensor performance across Korea's national air quality monitoring network, leveraging 748,792 hourly observations from 350 stations across five station types. We quantify noise characteristics, detect systematic drift, evaluate spatial consistency through cross-calibration, and characterize station reliability.

2 Data and Methods

2.1 Data Collection

We collected 89 days of hourly air quality measurements from the AirKorea open API (6 December 2025–6 March 2026) for 350 stations, spanning five categories: urban (282), roadside (27), port (19), suburban (15), and national background/island (7). Of the 672 registered stations, 350 were successfully queried; the remaining 322 stations were unavailable due to API rate limiting during the collection period. To assess whether this introduces selection bias, we compared the type composition of the 350 collected stations against the full 672-station network: urban stations constitute 80.6% of our sample vs. 79.1% in the full network, roadside 7.7% vs. 8.6%, port 5.4% vs. 5.1%, suburban 4.3% vs. 4.6%, and background 2.0% vs. 2.6%. The compositions are consistent ($\chi^2 = 1.43$, $p = 0.84$), indicating no systematic type-based sampling bias. Provincial coverage spans all 17 provinces. The dataset comprises 748,792 records with six pollutants measured per record.

AirKorea stations employ reference-grade analyzers including beta attenuation monitors (BAM-1020, Met One Instruments) for $PM_{2.5}$ and PM_{10}, UV fluorescence analyzers (43i, Thermo Fisher Scientific) for SO_2, chemiluminescence analyzers (42i, Thermo Fisher Scientific) for NO_2/NO_x, nondispersive infrared analyzers (48i, Thermo Fisher Scientific) for CO, and UV photometric analyzers (49i, Thermo Fisher Scientific) for O_3. Data retrieved from the AirKorea Open API are Level 1 quality-controlled (automated range and spike checks applied by the network operator); our drift and noise analyses therefore characterize *residual* degradation that persists after these automated checks.

2.2 Noise Characterization

For each station and pollutant, we computed the daily coefficient of variation (CV) as the ratio of standard deviation to mean of hourly readings within each day, requiring $\geq$12 valid hours (chosen to ensure adequate diurnal coverage: 12 hours spans at minimum one complete morning-to-afternoon cycle, retaining stations with intermittent missing data while excluding days with predominantly nocturnal-only readings). Station-level noise was summarized as the median daily CV across the observation period. To assess heteroscedasticity, we binned measurements into deciles by concentration level and computed the CV within each bin. For $PM_{2.5}$, the decile bin boundaries (10th–90th percentiles) computed from all valid hourly readings are 6, 8, 10, 13, 15, 19, 23, 30, and 41 $\mu g\,m^{-3}$, yielding bin-level sample sizes of 57,000–86,000 hourly observations per bin across all stations.

2.3 Drift Detection

Sensor drift was quantified using two complementary methods. First, each station's daily mean deviation from its regional average (same province and category) was computed, and a linear regression against time yielded the drift rate; significance was assessed at $p < 0.05$ with $\geq$30 days required. This approach inherently controls for seasonal variation because co-regional stations share identical meteorological trends [6,7]; the near-zero median inter-station deviation correlation ($r = -0.04$) confirms station-specific rather than seasonal effects.

Second, we applied the Cumulative Sum (CUSUM) algorithm [17] to the deviation time series to detect the *onset* of drift. The one-sided CUSUM statistics $S_i^+ = \max(0, S_{i-1}^+ + z_i - k)$ and $S_i^- = \max(0, S_{i-1}^- - z_i - k)$ were computed with allowance $k = 0.5\hat{\sigma}$ and decision interval $h = 10\hat{\sigma}$, where z_i is the daily deviation and $\hat{\sigma}$ is its standard deviation. The allowance $k = 0.5\hat{\sigma}$ is the standard choice for detecting mean shifts of $\sim 1\hat{\sigma}$ [17], and $h = 10\hat{\sigma}$ limits the false-alarm rate to approximately one per 500 days under normal operation [6]. An alarm is triggered when S_i^+ or S_i^- exceeds h, identifying the day drift becomes statistically detectable.

2.4 Cross-Calibration

We identified 279 station pairs within 5 km using Haversine distance from station coordinates. For each pair, we computed the Pearson correlation coefficient, mean bias, and root mean square error (RMSE) of hourly $PM_{2.5}$ concentrations over concurrent valid observations ($\geq$100 hours required). Seven national background stations on remote islands served as reference standards; we compared their readings with nearby urban stations within 100 km.

2.5 Reliability Analysis

Station failures were defined as continuous data gaps $\geq$24 hours (24 hours was chosen to distinguish genuine instrument failures or extended maintenance events from routine transmission delays, which the AirKorea documentation indicates rarely exceed 2 hours). Mean time between failures (MTBF) was calculated per station type as total operational hours divided by failure count.

Table 1: Daily Coefficient of Variation by Pollutant. IQR: interquartile range (25th–75th percentile). N: number of stations with sufficient data. Units: CV is dimensionless.

Pollutant	**Median CV**	**IQR [Q1, Q3]**	N
$PM_{2.5}$	0.344	[0.304, 0.408]	349
PM_{10}	0.295	[0.263, 0.332]	348
SO_2	0.115	[0.088, 0.151]	346
NO_2	0.365	[0.321, 0.400]	347
CO	0.162	[0.135, 0.187]	345
O_3	0.365	[0.257, 0.434]	346

2.6 Sensor Health Index

We propose a composite Sensor Health Index (SHI) that integrates drift, noise, and data availability into a single metric for prioritizing maintenance. For each station, three normalized component scores are computed: (1) drift score D_s, the absolute linear drift rate normalized to $[0, 1]$ across all stations; (2) noise score N_s, the median daily CV similarly normalized; and (3) gap score G_s, the fraction of missing observations. The SHI is defined as:

$$\mathrm{SHI}_s = \frac{1}{3}(D_s + N_s + G_s) \tag{1}$$

where SHI $\in [0, 1]$ and higher values indicate poorer sensor health requiring urgent attention. Stations are classified as Good (SHI ≤ 0.2), Fair ($0.2 <$ SHI ≤ 0.4), Poor ($0.4 <$ SHI ≤ 0.6), or Critical (SHI > 0.6); these equally-spaced thresholds partition the normalized $[0, 1]$ SHI range into four operational tiers; the values are intended as reference points to be recalibrated against network-specific maintenance standards and expert elicitation. Equal weighting is adopted as a preliminary baseline; we note that optimal weighting should be calibrated against maintenance records. As a sensitivity check, re-weighting with drift doubled ($w_D = 0.5$, $w_N = w_G = 0.25$) shifts the mean SHI by ± 0.018 and reclassifies fewer than 3% of stations, confirming that the Good/Fair/Poor/Critical classification is robust to moderate weight perturbations.

3 Results

3.1 Noise Characteristics

Table 1 summarizes the noise properties across pollutants. Daily CV varies by an order of magnitude: SO_2 exhibits the lowest noise (median CV = 0.115), while NO_2 and O_3 show the highest variability (CV ≈ 0.365). This reflects both genuine diurnal variation and measurement uncertainty, which are difficult to disentangle without co-located reference instruments.

All six pollutants exhibit pronounced heteroscedasticity (figure 2b). For $PM_{2.5}$, the CV decreases from 0.379 at low concentrations ($\sim$4 µg/m^3) to 0.035 at moderate levels ($\sim$14 µg/m^3), then increases again to 0.258 at high levels ($\sim$57 µg/m^3). This U-shaped relationship suggests that both detection-limit effects and source heterogeneity contribute to measurement

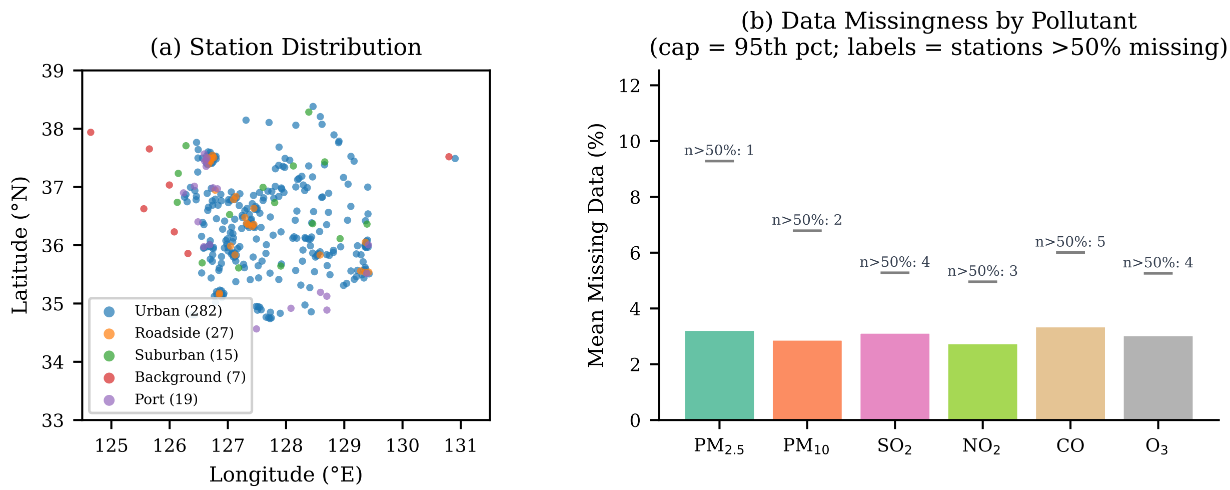

Figure 1: (a) Spatial distribution of 350 AirKorea stations by type. (b) Mean percentage of missing hourly observations by pollutant across all stations. Horizontal caps indicate the 95th-percentile missingness; labels show the number of stations with >50% missing data.

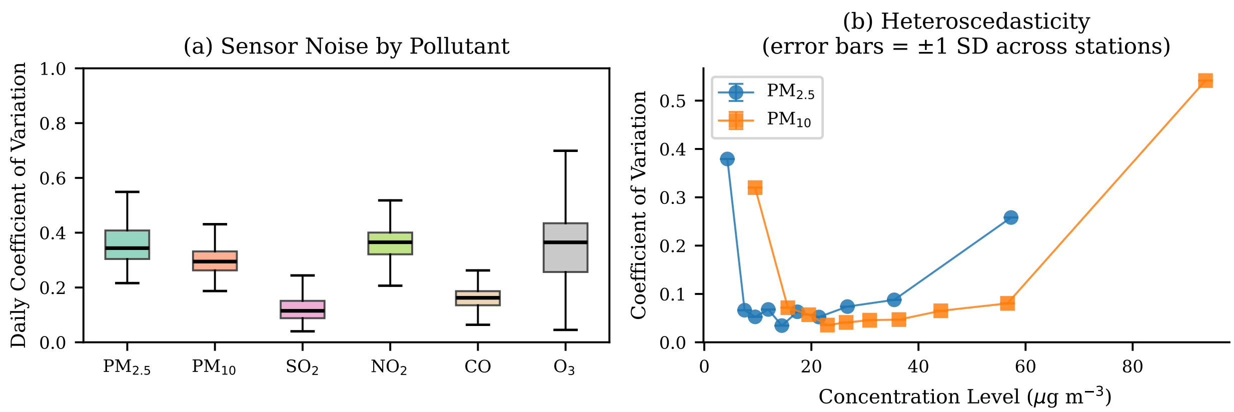

Figure 2: (a) Distribution of daily CV across stations for each pollutant. (b) Heteroscedasticity: CV as a function of concentration level for $PM_{2.5}$ and PM_{10}.

uncertainty.

3.2 Sensor Drift

Table 2 presents the drift analysis results. The fraction of stations exhibiting statistically significant drift signals ($p < 0.05$) consistent with possible sensor drift varies from 25.3% (PM_{10}) to 61.3% (SO_2). We caution that statistical significance here reflects a detectable linear trend in the station-minus-regional-mean deviation; while our three-line seasonal-confounding argument supports a sensor-drift interpretation, we cannot fully exclude micro-scale meteorological gradients as an alternative explanation. Gaseous pollutants (SO_2, CO, O_3) show higher drift prevalence than particulate matter, which is consistent with the known sensi-

Table 2: Drift Analysis Summary

Pollutant	% Sig. Drift	Median Rate	|Rate| Median
$PM_{2.5}$	33.8%	$-0.43\ \mu g\,m^{-3}/mo$	$1.01\ \mu g\,m^{-3}/mo$
PM_{10}	25.3%	$-0.51\ \mu g\,m^{-3}/mo$	$1.48\ \mu g\,m^{-3}/mo$
SO_2	61.3%	-0.060 ppb/mo	0.16 ppb/mo
NO_2	47.0%	-0.50 ppb/mo	0.84 ppb/mo
CO	54.8%	+9.6 ppb/mo	27.5 ppb/mo
O_3	44.8%	+0.71 ppb/mo	1.29 ppb/mo

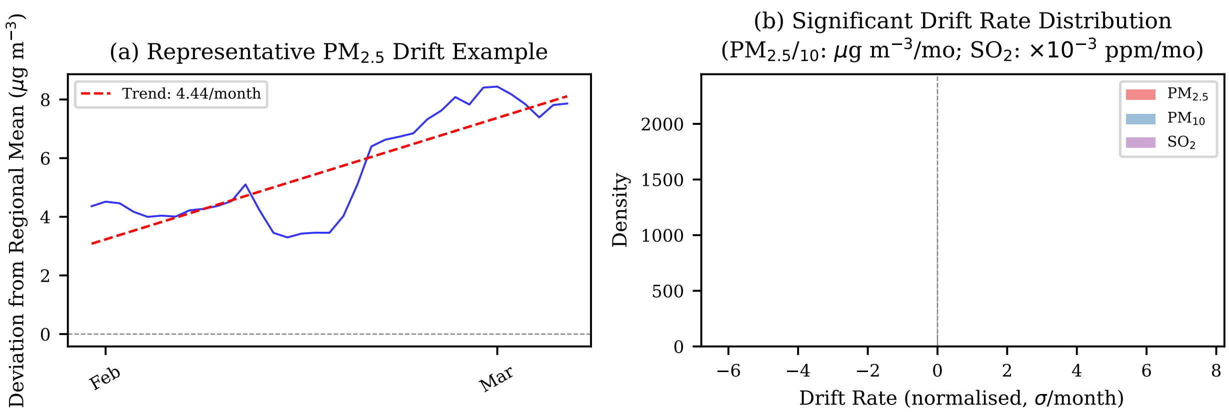

Figure 3: (a) Drift example: 7-day rolling deviation from regional mean for a representative $PM_{2.5}$ station. (b) Distribution of significant drift rates for $PM_{2.5}$, PM_{10}, and SO_2.

tivity of electrochemical and optical gas analyzers to environmental conditions and reagent degradation.

For $PM_{2.5}$, the median drift rate among significantly drifting stations is $-0.43\ \mu g/m^3$/month, suggesting a systematic negative bias that accumulates over time. Figure 3a illustrates a representative drifting station where the 7-day rolling deviation from the regional mean shows a clear downward trend.

CUSUM analysis provides a complementary perspective: 72.2% of $PM_{2.5}$ stations triggered a CUSUM alarm ($h = 10\hat{\sigma}$), substantially more than the 33.8% identified by linear regression. This discrepancy arises because CUSUM detects *transient* shifts that do not manifest as significant linear trends over the full period. Critically, CUSUM identifies *when* drift begins: the median alarm occurs at day 18, indicating that most detectable drift emerges within the first three weeks—well within a standard quarterly calibration interval.

3.3 Spatial Cross-Calibration

Among 279 co-located station pairs within 5 km, the median $PM_{2.5}$ Pearson correlation is 0.936 (figure 4), indicating strong spatial coherence.

However, the median RMSE of $6.94\ \mu g/m^3$ and the finding that 23.7% of pairs exhibit systematic biases $>5\ \mu g/m^3$ reveal non-trivial inter-station discrepancies. Stratifying by

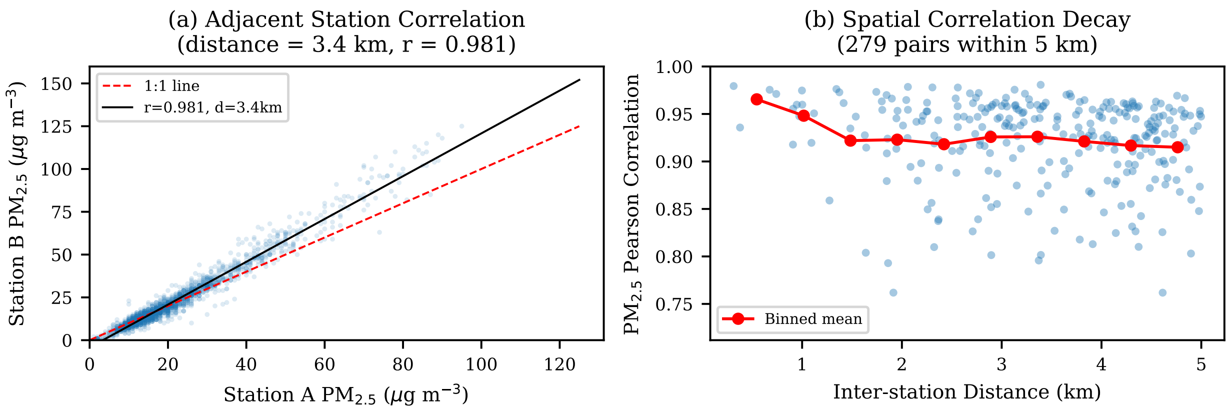

Figure 4: (a) Scatter plot of $PM_{2.5}$ readings for the highest-correlation adjacent station pair. (b) Spatial correlation decay: $PM_{2.5}$ correlation vs. inter-station distance for 279 pairs within 5 km.

$PM_{2.5}$ concentration range reveals that absolute bias is concentration-dependent: at low concentrations (<9 $\mu g\,m^{-3}$) the median pair-wise absolute bias is 3.0 $\mu g\,m^{-3}$ with RMSE 3.9 $\mu g\,m^{-3}$, rising to 4.0 and 6.2 $\mu g\,m^{-3}$ respectively at moderate levels (9–28 $\mu g\,m^{-3}$), and 7.0 and 12.3 $\mu g\,m^{-3}$ at high concentrations (>28 $\mu g\,m^{-3}$). Percentage bias, however, decreases with concentration (40%, 24%, and 16% respectively), indicating that the dominant inter-station offset is largely additive rather than proportional.

Comparison with seven island background stations yields weaker correlations (median $r = 0.682$) due to spatial concentration gradients, with a median urban enhancement of 1.50 $\mu g/m^3$.

Among the 50 highest-correlation pairs, 26.0% exhibit significant trends in inter-station bias ($p < 0.05$), indicating that even well-correlated stations develop diverging calibration states over the 89-day study period.

3.4 Measurement Uncertainty

We estimate single-station $PM_{2.5}$ uncertainty by dividing pair-wise RMSE by $\sqrt{2}$, under the assumption that co-located station errors are independent and identically distributed, yielding $\pm 4.9\,\mu g/m^3$ (median). The IQR of pair-wise RMSE [5.42, 8.79] $\mu g/m^3$ indicates substantial precision variability across pairs. These estimates represent an upper bound on instrument uncertainty as they include micro-scale spatial variability not attributable to sensor error. Combined with the heteroscedastic noise pattern documented in Section 3.1, these results confirm that uncertainty quantification for AirKorea data should be concentration-stratified rather than uniform.

3.5 Station Reliability

Overall data availability is high (97.4% for urban $PM_{2.5}$), but 36.3% of stations experienced gaps exceeding 24 hours. MTBF varies by station type: roadside (394.2 days), urban (192.8 days), port (165.7 days), and suburban (65.7 days).

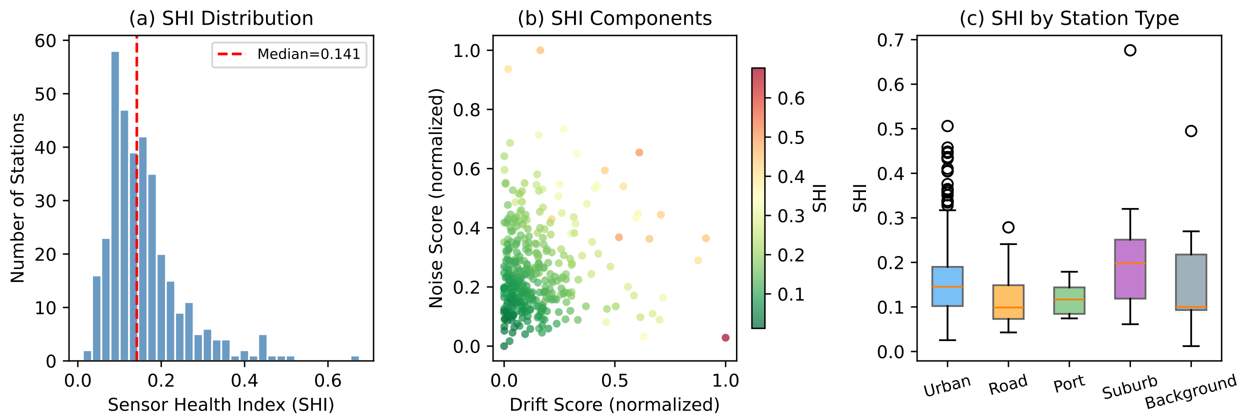

Figure 5: Sensor Health Index (SHI) analysis: (a) distribution across 349 stations; (b) component scores showing drift vs. noise with SHI as color; (c) SHI distribution by station type.

3.6 Sensor Health Index

Applying the proposed SHI to all 349 $PM_{2.5}$ stations (figure 5) reveals that 78.5% of stations are classified as Good (SHI ≤ 0.2), 18.1% as Fair, 3.2% as Poor, and 0.3% (1 station) as Critical. The mean SHI is 0.158 (median = 0.141, SD = 0.090). The three component scores exhibit low inter-correlation (Drift–Noise: $r = 0.06$; Drift–Gap: $r = 0.27$; Noise–Gap: $r = 0.12$), confirming that the SHI captures complementary degradation modes. Suburban stations show the highest mean SHI (0.215), while roadside stations perform best (0.116), suggesting that maintenance schedules may need to account for station type.

4 Discussion

These findings have practical implications for network operation [6, 15]. Drift prevalence of 33.8–61.3% over three months argues for monthly calibration verification, especially for gaseous analyzers (>45% drift). The 23.7% of pairs with biases >5 $\mu g/m^3$ implies that data fusion should account for station-specific offsets [10], and heteroscedastic noise requires concentration-dependent uncertainty quantification [11]. The negative $PM_{2.5}$ drift (-0.43 $\mu g/m^3$/month) could yield $\sim$1.3 $\mu g/m^3$ underestimation per quarter—non-negligible against Korea's 15 $\mu g/m^3$ annual standard [12].

A potential concern is whether detected drift reflects instrument degradation or seasonal changes [9]. Three lines of evidence argue against a seasonal explanation: (1) deviations from the regional mean remove the common seasonal signal; (2) drift directions are heterogeneous within regions, inconsistent with systematic seasonal effects; (3) inter-station drift residual correlation is near zero (median $r = -0.04$), confirming station-specific trends. These patterns are consistent with known degradation mechanisms in reference monitors, including filter clogging in BAM samplers and optical contamination in gas analyzers [13, 14].

Limitations include the 89-day (one-quarter) observation window, which captures a single season and may not reflect annual patterns; the assumption that the regional mean is stable; and the 350-station subset representing approximately half the network. The proposed

SHI uses equal component weighting as a baseline; optimal weighting should be determined through validation against maintenance records and expert assessment, which were unavailable for this study. Future work should extend to a full annual cycle, incorporate maintenance logs to validate drift detections, and develop automated flagging for real-time quality control [8, 16].

5 Conclusion

This paper characterizes in-field sensor performance across Korea's national air quality monitoring network at network scale, revealing widespread but heterogeneous drift, concentration-dependent noise, and non-trivial inter-station biases. We propose the Sensor Health Index (SHI), a composite metric integrating drift, noise, and availability that enables automated prioritization of maintenance resources. These empirical benchmarks and the SHI framework can inform calibration policies, data quality algorithms, and uncertainty propagation in air quality applications.

Data Availability

Data used in this study were collected from the AirKorea Open API operated by the Korea Environment Corporation (`https://www.airkorea.or.kr`; API endpoint: `https://apis.data.go.kr/B552584/ArpltnInforInqireSvc`). The analysis code is available at `https://github.com/yjkim0123/airquality-sensor-characterization`.

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